

LIGHTING & SMALL POWER DESIGN REPORT

Project	Duplex
Client	—
Prepared by	AutoDesign
Date	2026-07-10
Standard	BS7671 / EN 12464-1
Supply	230V, 50Hz, 3-phase

1. Lighting Design Calculations

Fixture layout and counts determined by the Lumen (Zonal Cavity) method: $N = (E \times A) / (F \times UF \times MF)$ where E = required illuminance (lux), A = room area (m²), F = fixture flux (lm), UF = utilisation factor, MF = 0.80.

Room	Type	Area (m²)	Req'd Lux	Room Index	UF	Fixture Flux (lm)	N Fixtures	Achieved Lux
kitchen	kitchen	14.5	300.0	0.71	0.44	10000.0	1	243.8

2. Fixture Schedule

Room	Fixture Type	Qty	Wattage (W)	Total Load (W)
kitchen	FloodLight	1	120.0	120.0
TOTAL		1		

3. Distribution Board Schedule — Ground Floor

DB location: (-13.31 m, -0.66 m) from drawing origin. Incomer: 20 A MCB, 1.5 mm² cable.

Circuit	Description	Pts	Load/pt (W)	Load Factor	Phase Load (W)	V	I (A)	I+25% (A)	MCB (A)	Cable (mm²)
CCT1	LP-R1-01	1	120	0.85	102	230	0.52	0.65	6	1.0
CCT4	SP-R4-01, SP-R4-02,	6	150	1.0	900	230	4.60	5.75	6	1.0
CCT7	SK-R7-01, SK-R7-02	2	1060	1.0	2120	230	10.84	13.55	16	1.0

Normal running load: R = 3122 W Y = 0 W B = 0 W

Total power: 3122 W at 230 V Full load current: 13.57 A +30% spare: 17.65 A Selected incomer MCB: 20 A

4. Cable Selection Summary

Cable sizes selected per BS7671 Appendix 4, maximum 3% voltage drop (6.9 V at 230 V). Run lengths estimated using Manhattan-distance routing with 1.2× routing factor. All cables: 3-core (L+N+E) PVC/SWA or PVC/PVC twin-and-earth as appropriate.

Circuit	Type	Phase	Cable (mm²)	MCB (A)	VP (V)
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CCT1	Lighting	R	1.0	6	0.190	0.08%
CCT4	Socket	R	1.0	6	2.240	0.97%
CCT7	Socket Key	R	1.0	16	3.9	1.70%

5. Design Basis & Assumptions

- Illuminance targets: EN 12464-1 (offices 500 lux, bedrooms 150 lux, kitchens 300 lux, corridors 100 lux).
- Lumen method: maintenance factor MF = 0.80, utilisation factor from simplified zonal cavity table.
- Supply voltage: 230 V, 50 Hz single-phase, 3-phase incomer.
- Cable volt-drop limit: 3% of 230 V = 6.9 V (BS7671 Appendix 12).
- Protective device: MCB to BS EN 60898, next standard rating above 125% of design current.
- Socket-outlet quantities: BS7671 guidance; minimum one per 8 m² for living/kitchen areas.
- DB spare capacity: 30% above calculated full-load current for future load growth.
- Cable routing factor: 1.20 applied to straight-line distances for bends and set-downs.
- This report is produced by automated calculation and must be verified by a qualified engineer.